Department of Computer Science

BSCCS Final Year Project 2025-2026
Interim Report I

25CS017

Secure Offline P2P Acoustic Proximity Authentication between Smartphone and Laptop

(Volume ___ of ___)

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1 Introduction

1.1 Motivation & Background Information

The widespread usage of electronic devices has become a definition of modern society. Securing access to these devices is a fundamental challenge, as their compromise can lead to breaches of personal privacy, financial loss, and critical infrastructure failure. The scale of this importance is reflected in the investments in the cybersecurity market, which is projected to grow from USD 218.98 billion in 2025 to USD 562.77 billion by 2030 (Fortune Business Insights, 2025). In response to this universal need for protection, a vast ecosystem of authentication methods has been developed.

A significant limitation of many modern authentication systems is their dependence on an online network connection for validation, such as contacting a central server for two-factor authentication (2FA). As Gurevich et al. (2022) state, "the problem of secure communication and authentication without online servers is fundamental" (p. 1). This paper, therefore, focuses specifically on authentication methods that can be implemented securely in an offline environment, where devices must establish trust without any network connectivity.

1.2 Problem Statement:

The main problem is that it is impossible to guarantee complete security with any authentication method. Therefore, it is crucial to develop layered security mechanisms where the corruption of one layer does not lead to the corruption of other layers (Grassi et al., 2020; Ometov et al., 2018). For instance, current offline authentication technologies represent a fundamental trade-off between security, accessibility, and implementation cost. NFC offers strong security but requires specialized hardware not available across all devices. Bluetooth offers broader compatibility but suffers from relay attacks due to its long range. QR codes ensure universal accessibility but lack bidirectional communication for secure mutual authentication.

This creates gaps in the security system, particularly for RF-restricted and offline environments where radio frequency methods may be compromised, jammed, or prohibited.

1.3 Proposed Solution:

This research proposes a new acoustic-based authentication system that addresses these gaps by providing an additional security layer for , proximity-based authentication application, in RF-restricted and offline environments.

1.4 Project Aims

This project aims to use standard audio hardware on laptops and phones as a transport layer for a secure authentication protocol, replacing radio frequency or optical layers in scenarios where those methods are inefficient. The main objective is to evaluate the feasibility of acoustic authentication in terms of data transmission reliability and resistance to common attacks.

1.5 Project Scope

- A Python application for the laptop to manage challenges, responses, and access control.
- An iOS application to capture, decode, and respond to acoustic challenges.
- A functional demonstration where successful acoustic authentication unlocks a designated folder on the laptop.

2 Literature Review

Authentication technologies can be categorized by their transmission medium, each with distinct trade-offs.

2.1 Radio Frequency (RF) Transmission:

2.1.1 NFC

Near Field Communication (NFC) provides secure short-range communication (<10 cm) and supports strong cryptography. According to Ravilia et al. (2024), NFC tokens use cryptographic-response protocols, making the technology highly secure against threats like phishing and man-in-the-middle (MITM) attacks. In addition, Ravilia et al. (2024) suggest that NFC is more convenient than alternatives as it requires only a simple "tap" from the user.

While it can be used as a two-factor authenticator, which will be proximity-based and secure, NFC-based two-factor authentication has its own issues. For instance, as Heartfield and Loukas (2018) state, NFC is expensive and not common in the majority of laptops and desktop technologies. Moreover, Francillon et al. (2019) demonstrate the possibility of relay attacks against short-range wireless protocols like NFC. They showed that the physical proximity requirement is not a guarantee of presence, as the authentication session can be relayed over a greater distance.

Although NFC is more secure against certain attacks like phishing and man-in-the-middle, and can create a smoother authentication process than acoustic-based authentication, audio components are more common. Therefore, the acoustic authentication method is more universally accessible.

2.1.2 Bluetooth

Bluetooth, which offers a greater range (up to 100 m), on the other hand, is more popular than NFC technology. Padgette et al. (2017) state that the widespread adoption of Bluetooth makes this technology a good base for the development of innovative cybersecurity applications. Bluetooth-

based solutions also offer a convenient way of authentication that is less cumbersome than alternatives with secure authentication solutions (Mare et al., 2014). The Bluetooth protocol is also evolving by adding more security measures. Modern BLE uses Secure Connections Pairing, a strong method that provides protection against passive eavesdropping (Antonioli et al., 2020). Despite the introduction of ECDH to the Bluetooth system does not help to safely solve relay attacks. The most significant security flaw of Bluetooth for authentication is its relatively long range(<100 m). This means that a device can be authenticated from different rooms within this radius. Ryan (2013), an attacker can relay the Bluetooth signals between two devices that are far apart, tricking them into believing they are in close proximity.

Acoustic authentication, despite not being a more popular technology than Bluetooth, solves its core weakness, which is long proximity. Sound waves attenuates rapidly and allow for precise time-of-flight measurements, acoustic systems can enforce close proximity**.** As noted by Huth et al. (2015), this property of sound decreases the chances of relay attacks by ensuring the device is close to the authenticator within centimeters, not meters, using physics.

2.2 Optical Transmission:

2.2.1 QR code:

QR codes are a widely adopted optical method for user authentication. The main advantage is that a QR code can contain a significant amount of data, making the authentication session efficient and reliable without a round-trips (Eminagaoglu et al., 2014). Tirfe et al. (2021) state that it is also resistant to man-in-the-middle (MITM) attacks.

On the other hand, one-way factor authentication can be considered a security flaw. Tirfe et al. (2021) suggest that this technology is vulnerable to remote session swapping attacks because it does not have mutual authentication. It also depends on the phone's camera, which can be broken, limiting secure authentication options.

Unlike the one-way visual channel of a QR code, an acoustic channel can easily be made bidirectional. This helps to establish a secure communication session resistant to remote attackers (Zhang et al., 2014).

2.3 Audio Transmission:

2.3.1 Acoustic authentification:

This method uses sound waves to transmit data and has two major flaws. The acoustic channel has a very low data rate compared to RF or optical links, making it unsuitable for large data transfers and potentially resulting in higher latency during the authentication handshake (Mehrabi et al., 2020). In addition, performance can be degraded by noise, which leads to packet loss. This necessitates robust error-correction and can reduce reliability in noisy real-world settings (Nandakumar et al., 2014).

Even though the acoustic channel is not reliable in noisy environments, the characteristics of sound waves can act as a preventive mechanism against relay attacks because sound waves attenuate rapidly and are blocked by walls. This short physical range (<1m) enforces close proximity, providing a natural defense against the long-distance relay attacks that are a critical flaw in Bluetooth-based systems (Jakobsson & Wetzel, 2001). While the small data capacity poses challenges, it is not problematic for this application since this project aims to transfer only the cipher keys, which are small in size and do not require large data transfers. In addition, unlike NFC or Bluetooth LE, it requires no specialized hardware, leveraging components present in virtually all smartphones and laptops. This makes it a universally accessible solution that functions completely offline (Schürmann & Sigg, 2011).

Therefore, despite the existence of NFC, Bluetooth, and QR code-based authentications that provide a good level of security and are already in use by world-leading companies, they can still have security or implementation flaws in the scenarios described above. Although the acoustic

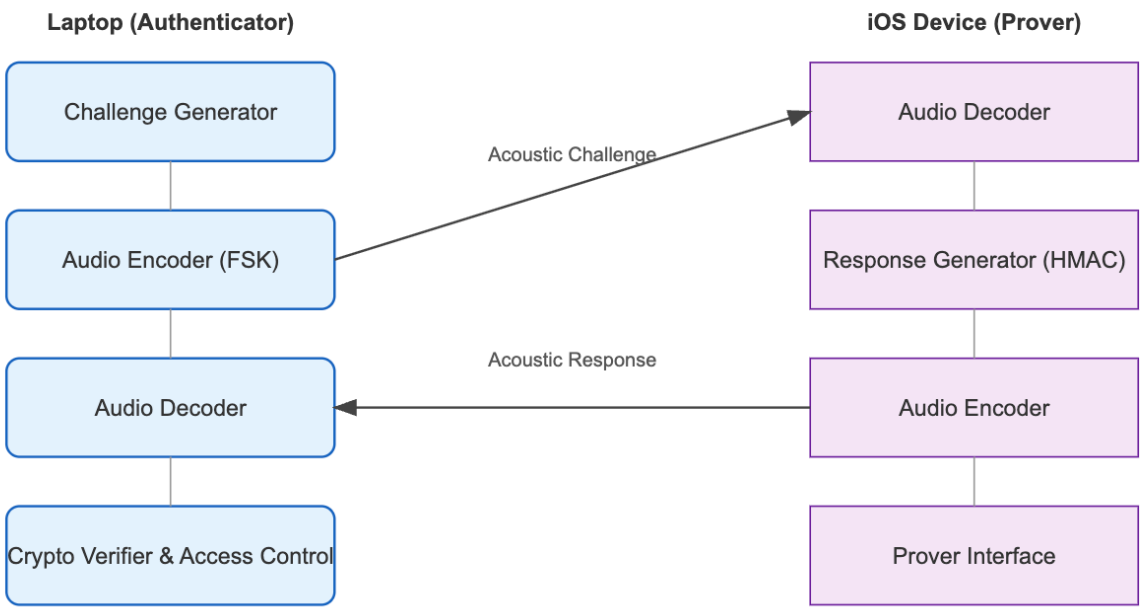
channel does not solve all the issues of other authentication methods, it can cover security needs in offline, RF-restricted environments that require close proximity between pairing devices.

3 Preliminary Design, Solution, System

3.1 System Overview

Acoustic authentication system will establish acoustic authentification channel between a Phone(iPhone) and Laptop(MacBook). This channel relies on per to peer communication channel, therefore it does not require connection to the Internet, or any type of 3 partyy technology. This desgn helps to provide secure authentification Offline and RF-restricted environment.

The proposed authentification system operates through challenge-response authentication over an acoustic channel. A laptop generates a cryptographic challenge, encodes it into an audible or near-ultrasonic audio signal, and sends using speaker. The iPhone receives this signal, decodes it, computes a cryptographic response using a shared secret, and transmits the response acoustically back to the laptop. The laptop validates the response to complete or deny authentication.



Acoustic-based Challenge–Response Authentication Flow

3.2 System Components

The proposed acoustic authentication solution is designed using a collection of standard hardware and custom software components. The system's building blocks can be logically divided into two main categories: the physical hardware that handles acoustic transmission and reception, and the software modules that manage the cryptographic and data-link protocols.

3.2.1 Hardware Components

The system is designed to operate on commodity hardware, requiring no specialized equipment. Its functionality relies on the standard audio input and output capabilities found on modern laptops and smartphones, which together form the physical layer for data transmission.

- **Laptop Speakers:** The built-in or external audio output system of the laptop. Its primary function is to transmit the acoustic challenge by playing the encoded audio tones.
- **Laptop Microphone:** The standard integrated microphone on the laptop. It is responsible for recording the incoming acoustic response signal sent from the mobile device.
- **iPhone Speaker:** The standard loudspeaker on the iOS device. It is used to send back the encoded acoustic response tone sequence to the laptop.
- **iPhone Microphone:** The built-in microphone on the iOS device. Its role is to capture the initial acoustic challenge from the laptop and forward the signal to the software decoding module.

3.2.2 Software Components

Complementing the hardware is a layered software stack that implements the core authentication logic. These modules handle everything from low-level signal processing to high-level cryptographic operations and user interaction, ensuring a robust and secure authentication process.

- **Audio Interface Module**: Manages all audio input/output operations using PyAudio (Python) and AudioKit (Swift), handling recording, playback, and synchronization at a 44.1 kHz sample rate.
- **Modulation & Demodulation Module:** Converts data between binary and acoustic signals using Frequency-Shift Keying (FSK) for transmission and employs FFT for frequency detection during reception.
- **Frame Construction Module:** Assembles data into structured frames with a Preamble, Header, Payload, and CRC checksum, managing the timing and structure of transmissions.
- **Challenge Generator:** Produces a cryptographically secure random nonce for each session using the secrets module (Python) or SecRandomCopyBytes (iOS).
- **Cryptographic Engine:** Computes and verifies HMAC-SHA256 hashes to ensure the integrity and authenticity of the challenge-response exchange.
- **Response Generator:** On the iOS device, computes the HMAC response to the received challenge and prepares it for acoustic transmission.
- **Verification and Access Control Module:** Compares the received response against the expected value and unlocks the secured resource upon a successful match.
- **Error Correction & Control Module:** Implements CRC-16 for error detection and manages retransmission protocols to ensure data reliability over the acoustic channel.
- **User Interface Layer (UI):** Provides the user-facing application on both devices, built with Swift UI for iOS and Tkinter/PyQt for the laptop

3.3 Algorithm

The core authentication mechanism employs a cryptographic challenge-response protocol that uses acoustic channel. The protocol ensures mutual authentication through shared secret verification while maintaining resistance to replay attacks

3.3.1 Challenge Generation:

The initiator (laptop) generates a cryptographic challenge using the following procedure:

C = os.urandom(16) # 128-bit random challenge

R_expected = HMAC(K, C, digestmod='sha256')

Algorithm Specification: The nonce C is generated using a cryptographically secure pseudorandom number generator (CSPRNG) as specified by Barker and Kelsey (2012). The 128-bit length provides sufficient entropy (2^128 possible values) to prevent brute-force attacks. The HMAC construction follows the standard by Krawczyk et al. (1997) using SHA-256 as the underlying hash function, providing 128-bit security strength against collision attacks.

3.3.2 Acoustic Encoding and Transmission:

The binary challenge C is converted to an acoustic signal using digital modulation techniques:

Frequency-Shift Keying (FSK) Implementation:

- Binary '0': Represented by frequency f_0 = 18000 Hz (near-ultrasonic)
- Binary '1': Represented by frequency f_1 = 19000 Hz
- Symbol duration: T_symbol = 50 ms
- Sampling rate: f_s = 44100 Hz

The modulated signal s(t) is constructed as:

$s(t) = A \cdot \sin(2\pi f_0 t)$ for binary '0'

$s(t) = A \cdot \sin(2\pi f_1 t)$ for binary '1'

Technical Rationale: FSK modulation provides robustness against amplitude variations and selective fading (Proakis & Salehi, 2001). The near-ultrasonic frequency range (18-19 kHz) minimizes audible disturbance while remaining within the frequency response of standard laptop speakers and smartphone microphones, typically capable of 16-22 kHz response (Michalevsky et al., 2014).

3.3.3 Signal Reception and Decoding:

The receiver (iOS device) processes the incoming acoustic signal through a demodulation pipeline:

Preprocessing Stages:

1. *Band-pass filtering*: 17-21 kHz Butterworth filter to remove out-of-band noise
2. *Normalization*: Automatic gain control to compensate for distance-dependent attenuation
3. *Frame synchronization*: Preamble detection using Barker code sequence
4. *FSK demodulation: Goertzel algorithm (Sysel & Rajmic, 2012) for efficient frequency detection*

The Goertzel algorithm computes the energy at frequencies f_0 and f_1:

$P_0 = |X(f_0)|^2$, $P_1 = |X(f_1)|^2$

Decision: binary '0' if $P_0 > P_1$, else binary '1'

3.3.4. Response Computation:

Upon successful decoding of challenge C, the prover computes:

text

R_device = HMAC(K, C)

using the same HMAC-SHA256 parameters as the authenticator. The shared secret K is securely stored in the device's keychain, protected by the iOS Secure Enclave.

3.3.5 Verification and Access Control:

The authenticator performs a constant-time comparison:

```
if secure_compare(R_device, R_expected):
    grant_access()
else:
    deny_access()
```

The secure_compare() function implements constant-time comparison to prevent timing attacks (Brumley & Boneh, 2005), ensuring the comparison time is independent of the number of matching bytes**Protocol Security Properties:**

- *Replay Protection*: Each challenge is used only once (nonce)
- *Forward Secrecy*: Compromise of K doesn't reveal past session keys
- *Mutual Authentication*: Both parties prove knowledge of K
- *Proximity Assurance*: Acoustic channel limits effective range to ~1 meter

3.4 Testing and Experimental Design

3.4.1 Testing Objectives

The testing phase aims to evaluate:

- **Functional accuracy** – correctness of authentication responses.
- **Signal reliability** – success rate of acoustic data transmission under varying noise conditions.
- **Security robustness** – resistance to replay, relay, and eavesdropping attacks.
- **Practical usability** – response time, error rate, and real-world feasibility.

3.4.2 Test Setup

Parameter	**Configuration**
Test Environment	Quiet indoor room (baseline), moderate noise (office), high noise (café)
Devices	Laptop (MacBook/Windows PC) & iPhone
Sampling Rate	44.1 kHz
Encoding Scheme	FSK (two-tone), adjustable frequency separation
Number of Trials	50 per environment

Parameter	Configuration
Distance Between Devices	10 cm – 100 cm

3.4.3 Testing Procedures

1. **Functional Validation**:
    - Run the full authentication sequence multiple times under ideal conditions.
    - Measure success rate and transmission latency.
2. **Noise Tolerance Tests**:
    - Introduce controlled background noise levels (e.g., white noise at 40–70 dB).
    - Measure decoding success rate and error rate.
3. **Security Tests**:
    - Attempt replay attacks by recording and re-transmitting previous response signals.
    - Attempt distance-based relay attacks (>1m) to confirm proximity enforcement.
4. **Performance Metrics**:
    - **Success Rate (%) = Successful Authentications ÷ Total Attempts**
    - **Average Latency (ms)** between challenge emission and authentication completion.
    - **Bit Error Rate (BER)** to assess transmission reliability.

4 Preliminary Design, Solution, System

4.1 Preliminary Testing Results

The initial two-month development phase has successfully resulted in a minimal viable prototype (MVP) that validates the core technical premise. Proof of concept demonstrated a complete, end-to-end authentication cycle under silent conditions. The system reliably transmits a cryptographic nonce from a laptop to an iOS device and subsequently receives and successfully verifies the correct HMAC-SHA256 response. This successful proof-of-concept, which culminates in a software-based "access granted" signal on the laptop, provides a stable and verified foundation for all subsequent development.

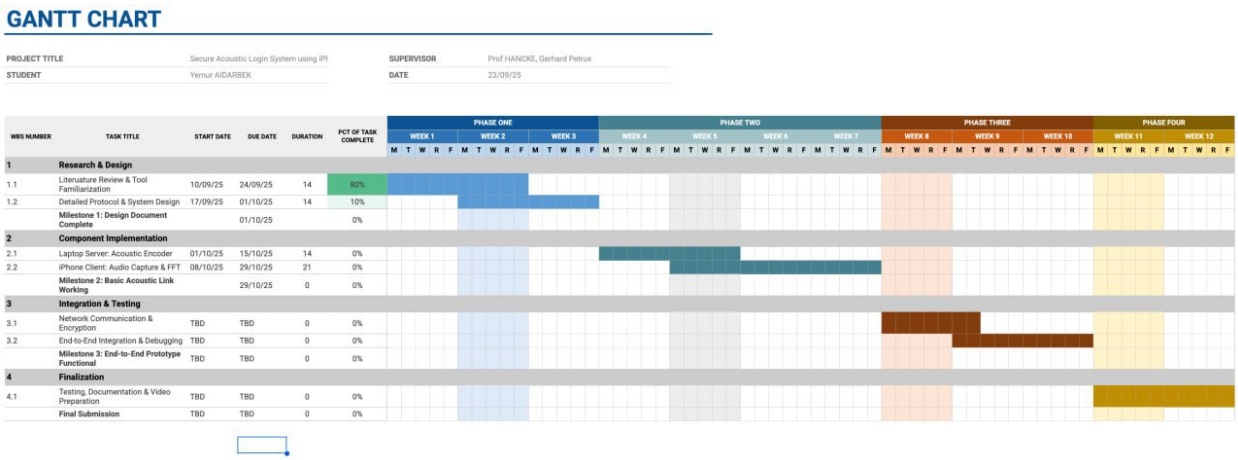
4.2 Future Improvement

Building upon the functional MVP, the next phase of development will focus on enhancing the system's robustness, security, and practical usability. Key areas for improvement include:

- **Implementing a Robust Data-Link Layer:** The current simple transmission will be replaced with a structured frame protocol featuring preambles for synchronization, payload length headers, and CRC checksums for error detection to significantly improve reliability.
- **Introducing Error Correction:** We will integrate forward error correction codes, such as Reed-Solomon, to correct bit errors without requiring retransmission, thereby increasing success rates in less-than-ideal acoustic environments.
- **Hardening Security:** The system will be fortified against replay attacks by implementing a secure nonce management system that logs used challenges. Furthermore, all comparison operations (e.g., HMAC verification) will be made constant-time to prevent timing-based side-channel attacks.

- **Developing a Basic User Interface:** A simple graphical user interface (GUI) will be developed for both the laptop and iOS application to replace command-line operations, providing users with clear status feedback and control over the authentication process.

4.3 Gantt Chart



GANTT CHART

PROJECT TITLE	Secure Acoustic Login System using iPI	SUPERVISOR	Prof HANCKE, Gerhard Petrus
STUDENT	Yernur AIDARBEK	DATE	23/09/25

WBS NUMBER	TASK TITLE	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
1	**Research & Design**				
1.1	Literuature Review & Tool Familiarization	10/09/25	24/09/25	14	80%
1.2	Detailed Protocol & System Design	17/09/25	01/10/25	14	10%
	Milestone 1: Design Document Complete		01/10/25		0%
2	**Component Implementation**				
2.1	Laptop Server: Acoustic Encoder	01/10/25	15/10/25	14	0%
2.2	iPhone Client: Audio Capture & FFT	08/10/25	29/10/25	21	0%
	Milestone 2: Basic Acoustic Link Working		29/10/25	0	0%
3	**Integration & Testing**				
3.1	Network Communication & Encryption	TBD	TBD	0	0%
3.2	End-to-End Integration & Debugging	TBD	TBD	0	0%
	Milestone 3: End-to-End Prototype Functional	TBD	TBD	0	0%
4	**Finalization**				
4.1	Testing, Documentation & Video Preparation	TBD	TBD	0	0%
	Final Submission	TBD	TBD	0	0%

Monthly Logs

Monthly Log (October 2025)

Current Time: Nov 10, 2025 (Mon) 11:24:53 PM

Submission Deadline: Oct 31, 2025 (Fri) 11:59:59 PM

Date Last Edited: Nov 10, 2025 (Mon) 11:24 PM

Content*:

Had Meeting with Professor

Completed Literature Review & Tool Familiarization

Completed System Design

Monthly Log (November 2025)

Current Time: Nov 10, 2025 (Mon) 11:28:28 PM

Submission Deadline: Nov 30, 2025 (Sun) 11:59:59 PM

Date Last Edited: Nov 10, 2025 (Mon) 11:28 PM

Content*:

- The system reliably transmits a cryptographic nonce from a laptop to an iOS device and subsequently receives and successfully verifies the correct HMAC-SHA256 response

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